Portes logiques

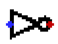
Table des matières

Portes logiques....1
1) La base des portes logiques :....3
1-1) NON (NOT) {inverseur} :....3
1-2) ET (AND) :....5
1-3) NON-ET (NAND) :....7
1-4) OU (OR) :....9
1-5) NON-OU (NOR) :....11
1-6) OU-EXCLUSIF (XOR) :....13
1-7) NON-OU-EXCLUSIF (XNOR) :....16
2) Portes logiques comportant plus de 2 entrées :....19
2-1) ET (AND) [3 à 4 entrées] :....19
2-2) NON-ET (NAND) [3 à 4 entrées] :....20
2-3) OU (OR) [3 à 4 entrées] :....21
2-4) NON-OU (NOR) [3 à 4 entrées] :....22
3) Portes logiques avec bus de données :....23
3-1) NON (NOT) {inverseur} :....23
3-2) ET (AND) :....24
1-3) NON-ET (NAND) :....25
3-4) OU (OR) :....26
3-5) NON-OU (NOR) :....27
3-6) OU-EXCLUSIF (XOR) :....28
3-7) NON-OU-EXCLUSIF (XNOR) :....29

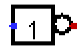
1) La base des portes logiques :

1-1) NON (NOT) {inverseur} :

Américain



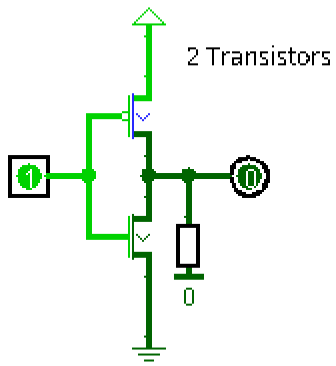
Europe



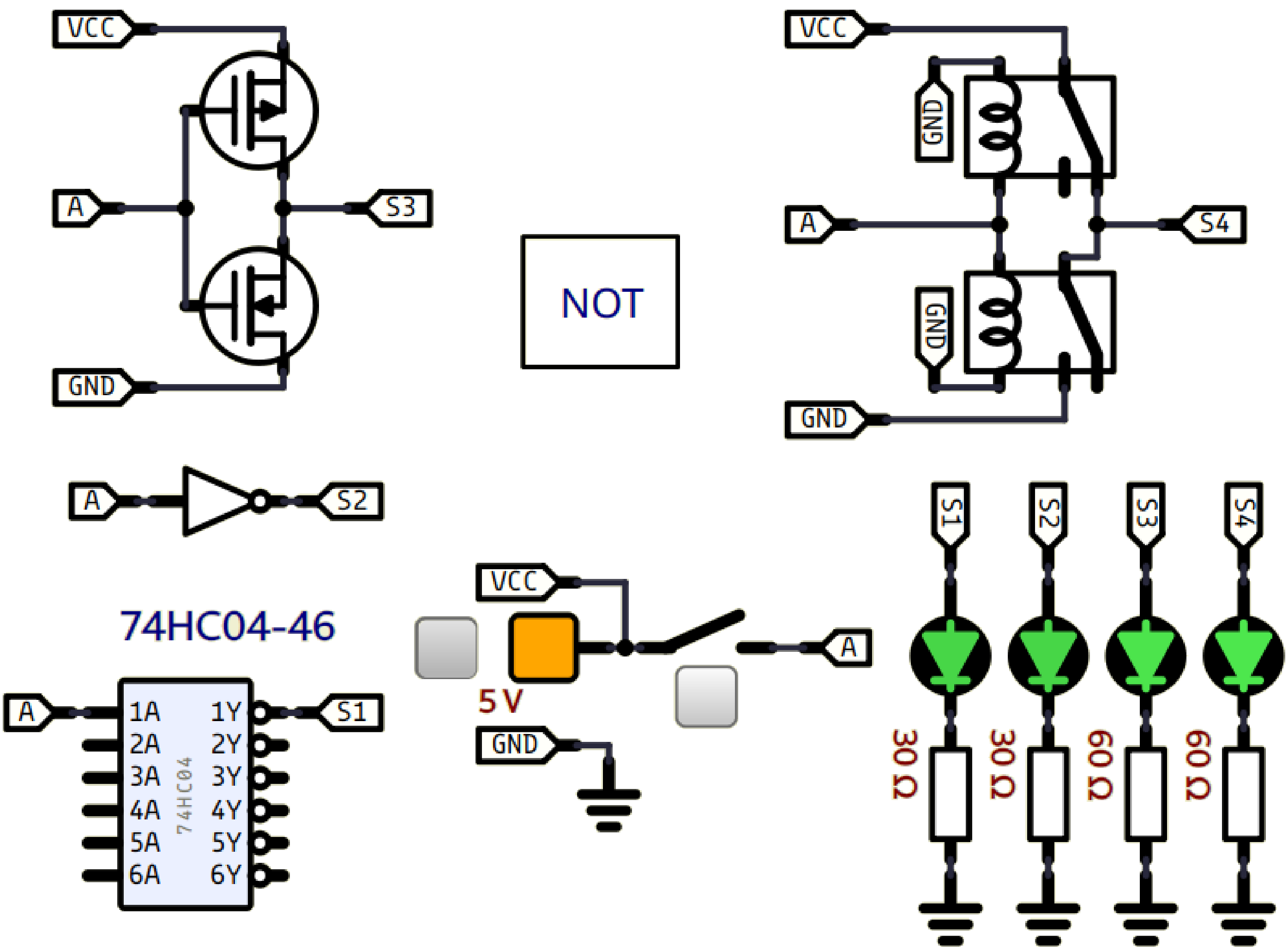
A	S
0	1
1	0

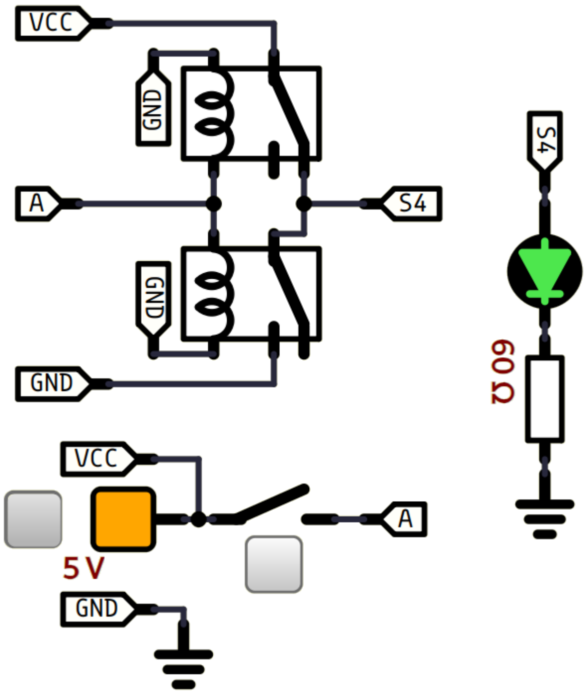
Transistors

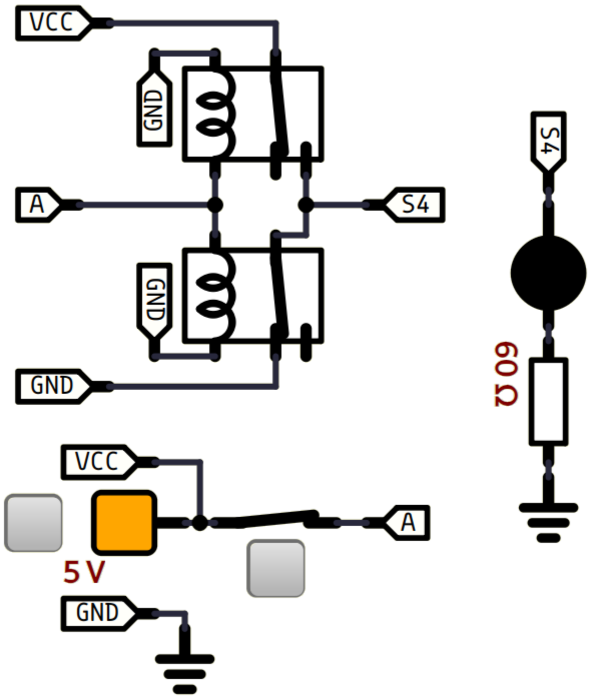
(logisim)



(simulide)




VCC
GND
A
S4
GND
GND
S4
60 Ω
VCC
5 V
A
GND

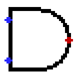

VCC
GND
A
S4
GND
GND
S4
60 Ω
VCC
5 V
A
GND

1-2) ET (AND) :

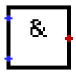
Transistors

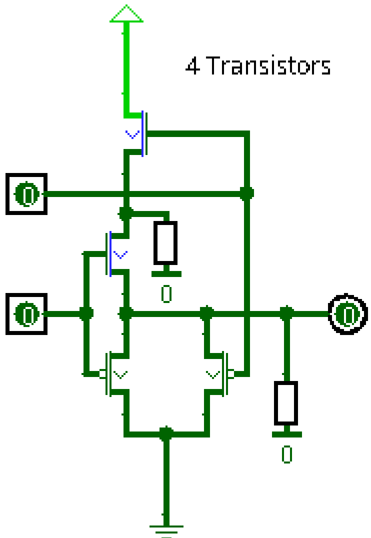
(logisim)

Américain



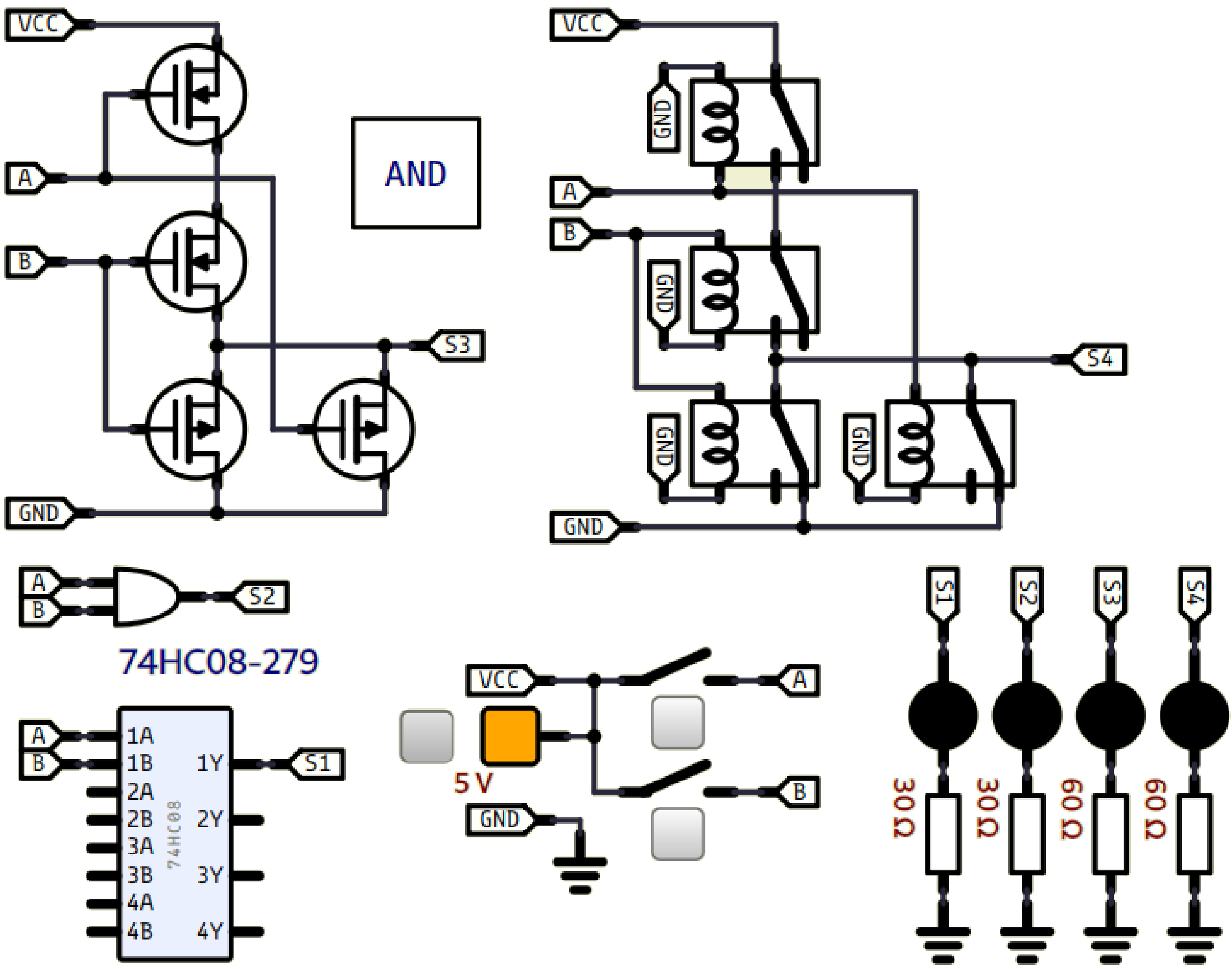
Europe

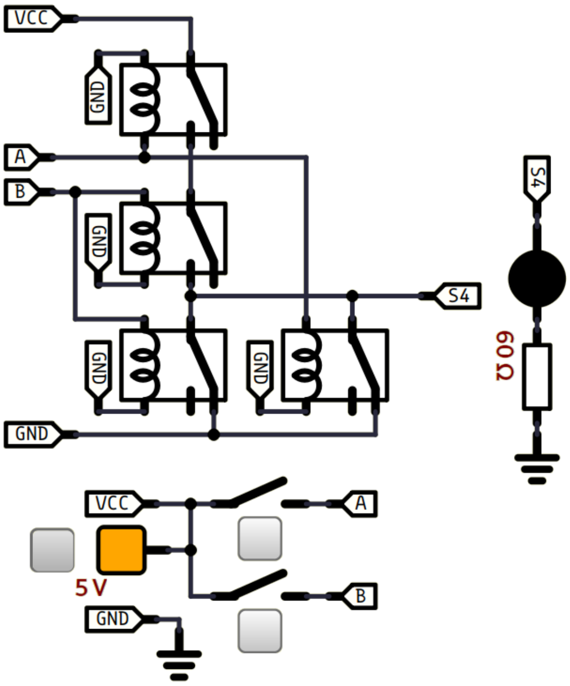


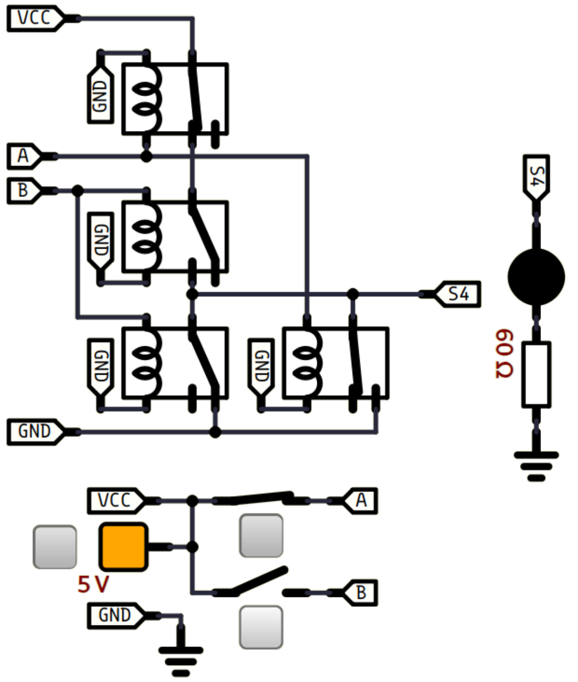


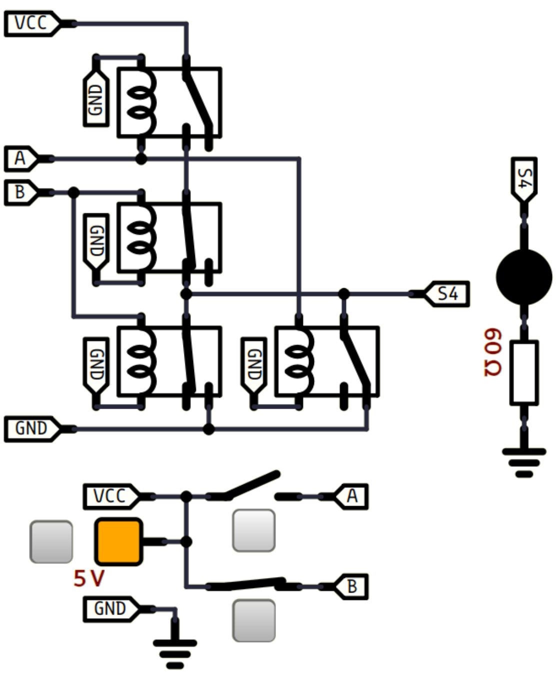
A	B	S
0	0	0
0	1	0
1	0	0
1	1	1

(simulide)




VCC
GND
A
B
S4
60 Ω
5 V


VCC
GND
A
B
S4
60 Ω
5 V


VCC
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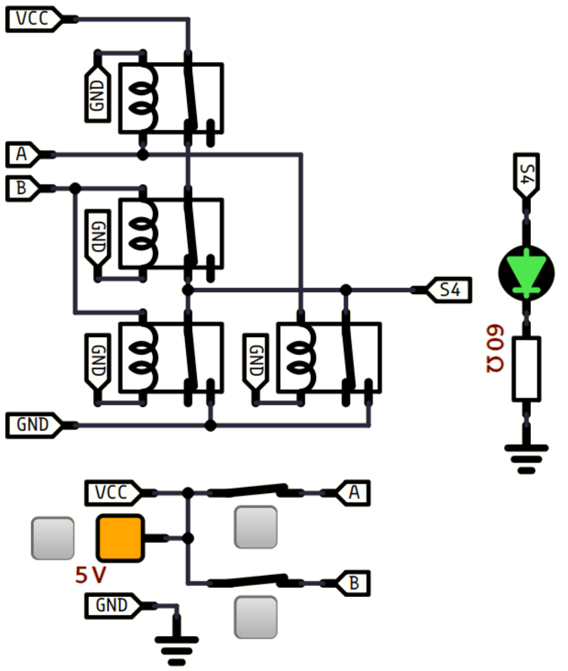
5 V

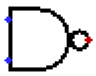

VCC
GND
A
B
S4
60 Ω
5 V

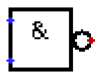
1-3) NON-ET (NAND) :

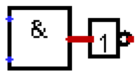
Transistors

(logisim)

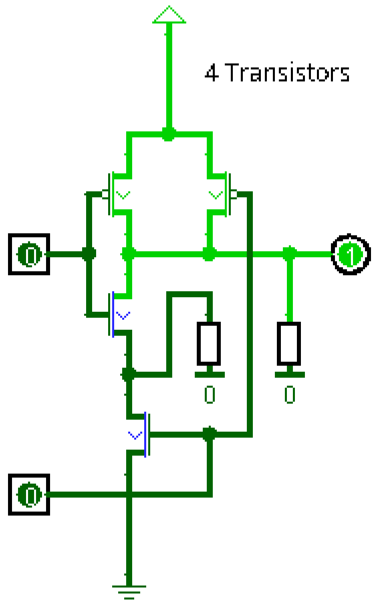
Américain Europe



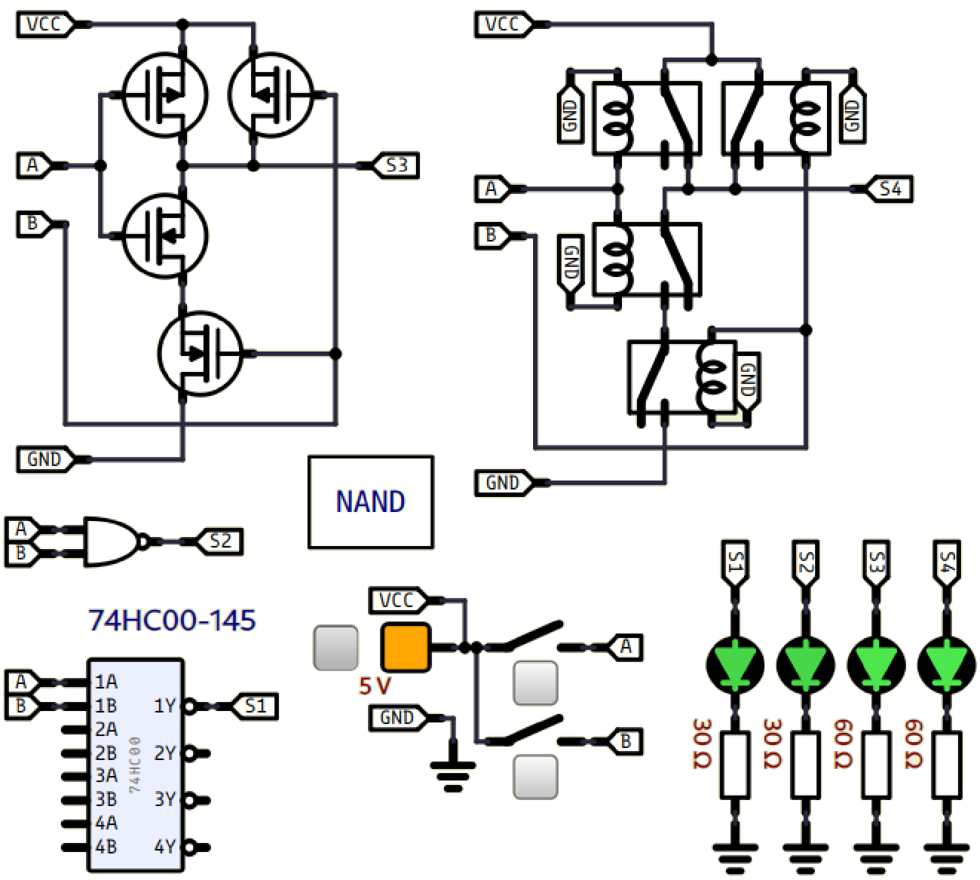


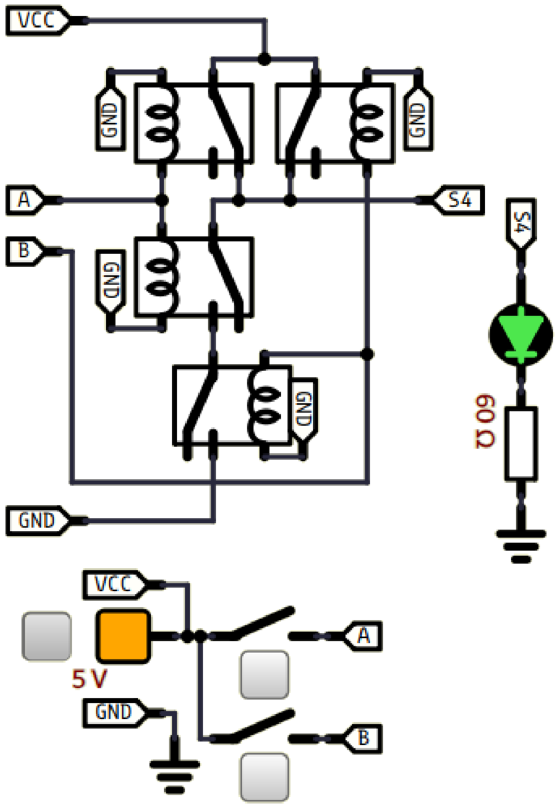


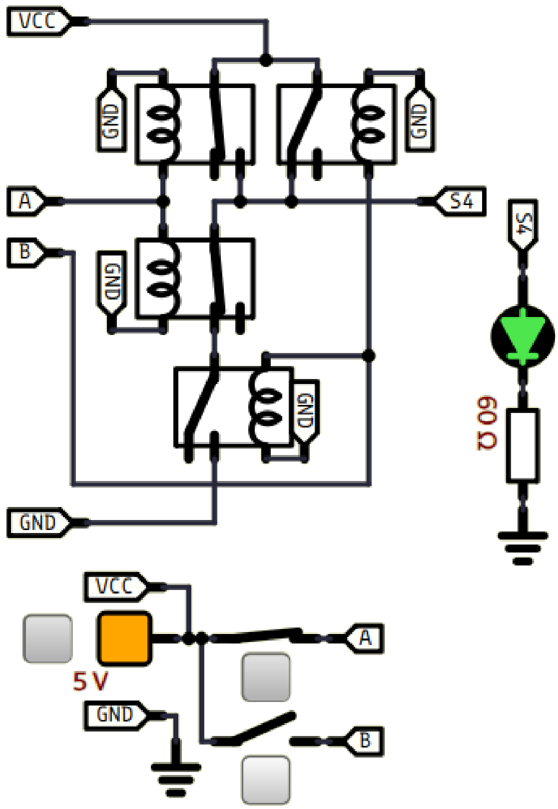
A	B	S
0	0	1
0	1	1
1	0	1
1	1	0

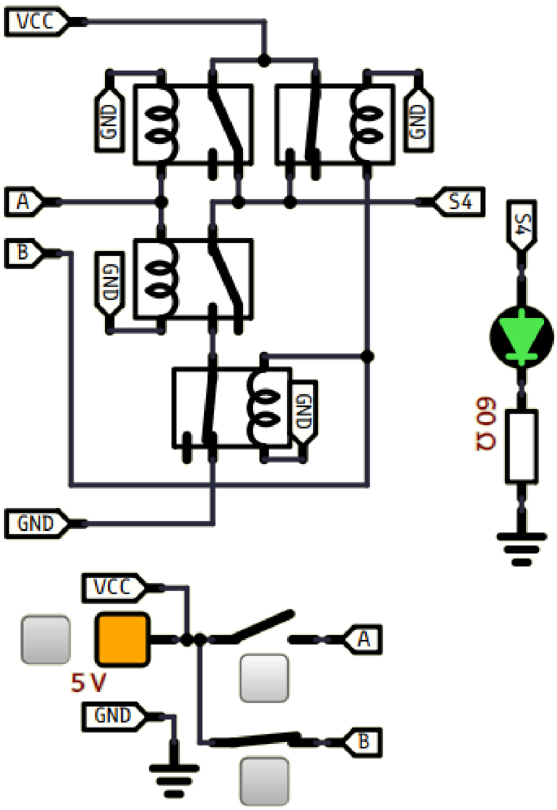


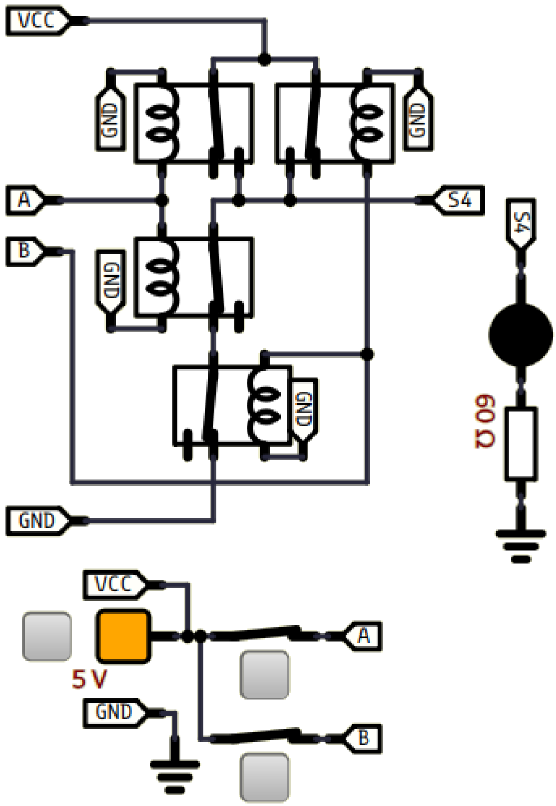
(simulide)




VCC
GND
A
B
S4
60 Ω
5 V


VCC
GND
A
B
S4
60 Ω
5 V


VCC
GND
A
B
S4
60 Ω
5 V

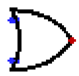

VCC
GND
A
B
S4
60 Ω
5 V

1-4) OU (OR) :

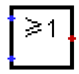
Transistors

(logisim)

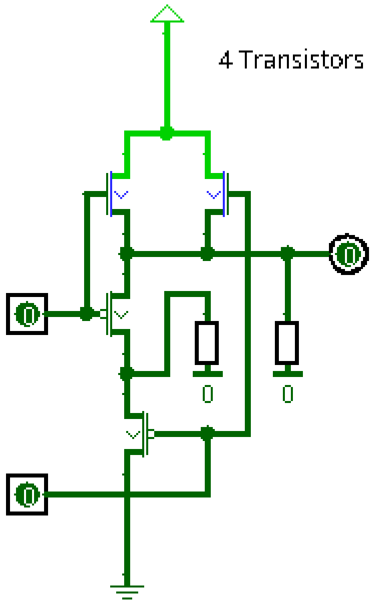
Américain



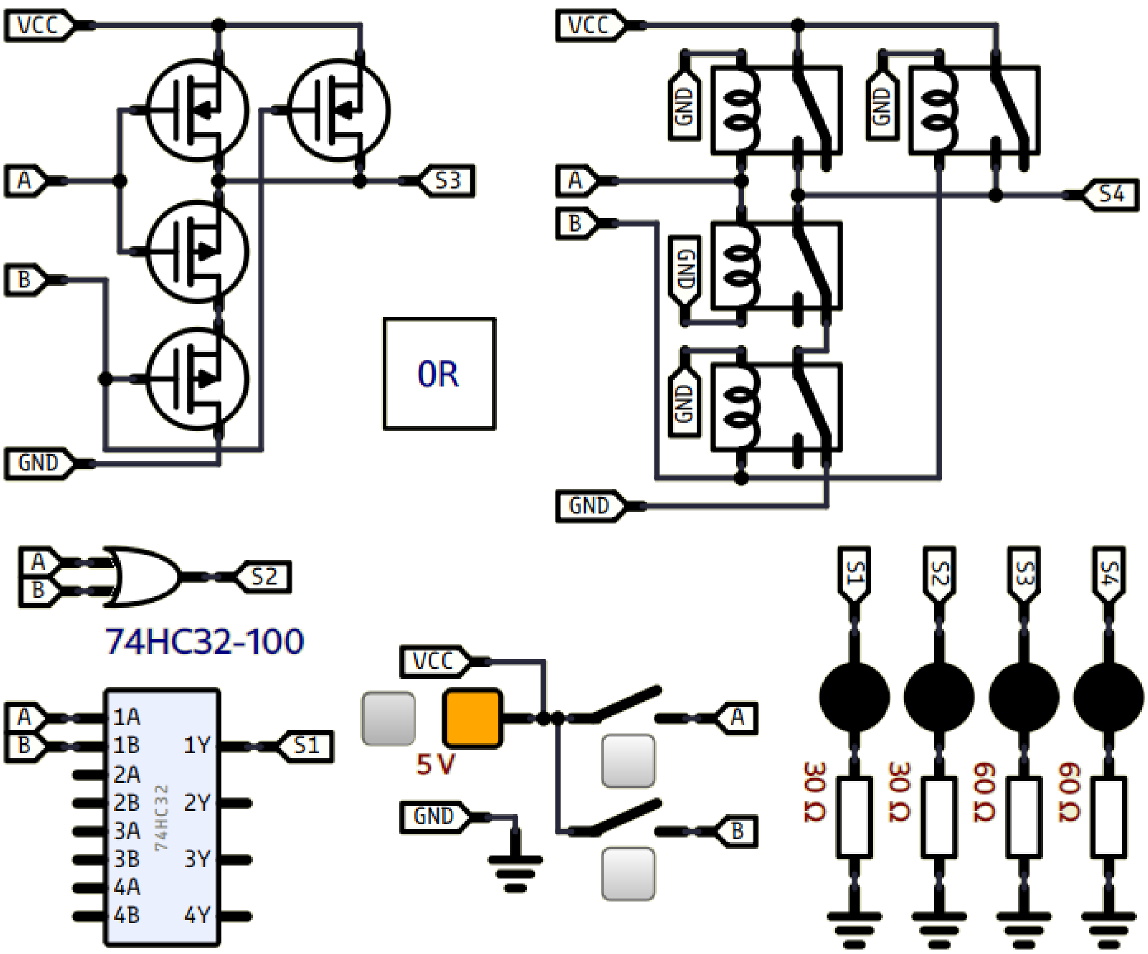
Europe

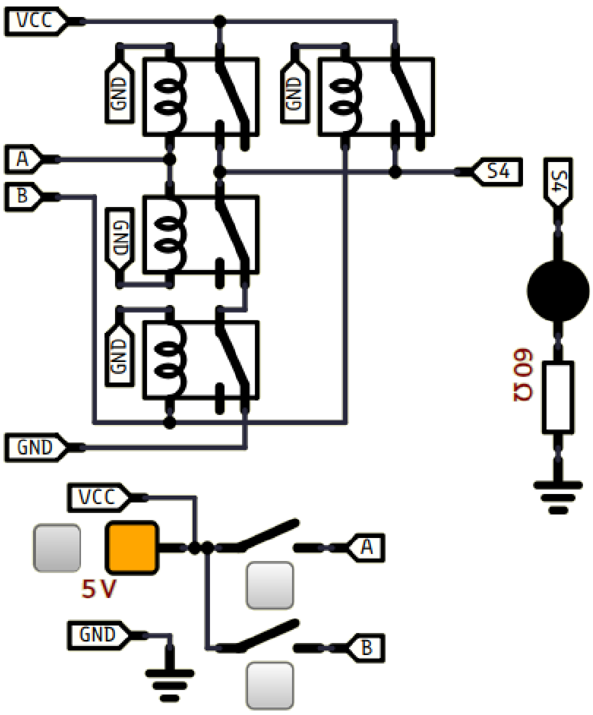


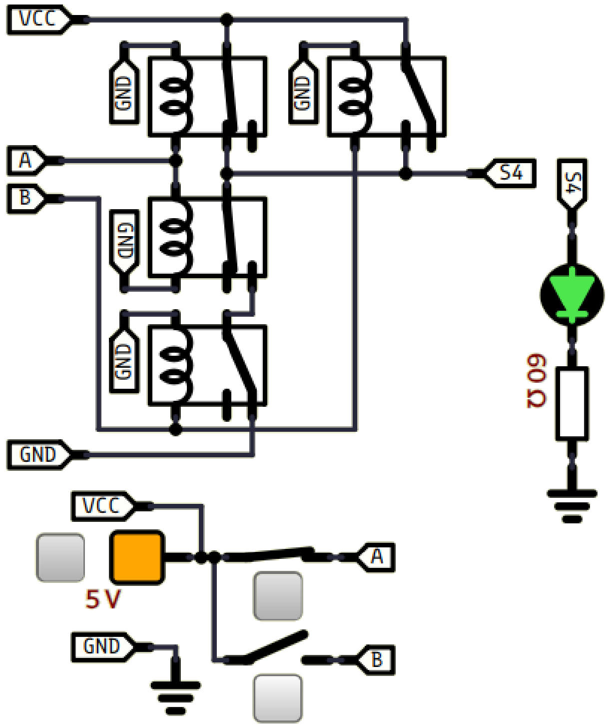
A	B	S
0	0	0
0	1	1
1	0	1
1	1	1

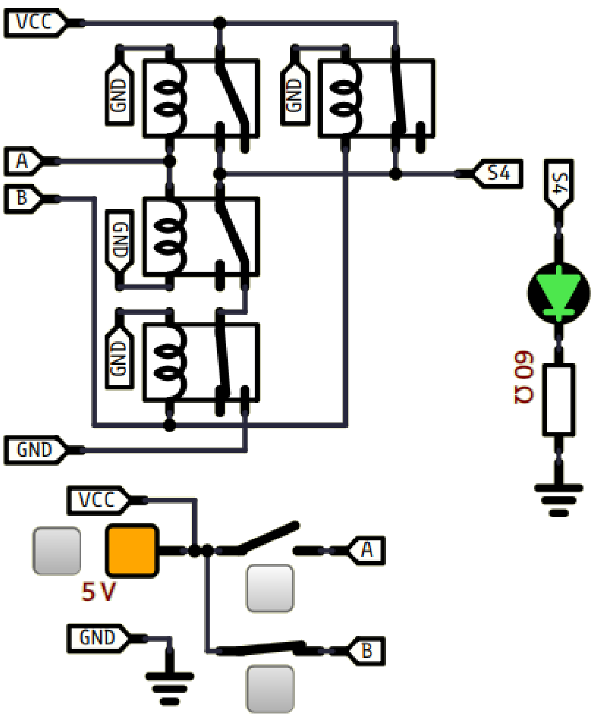


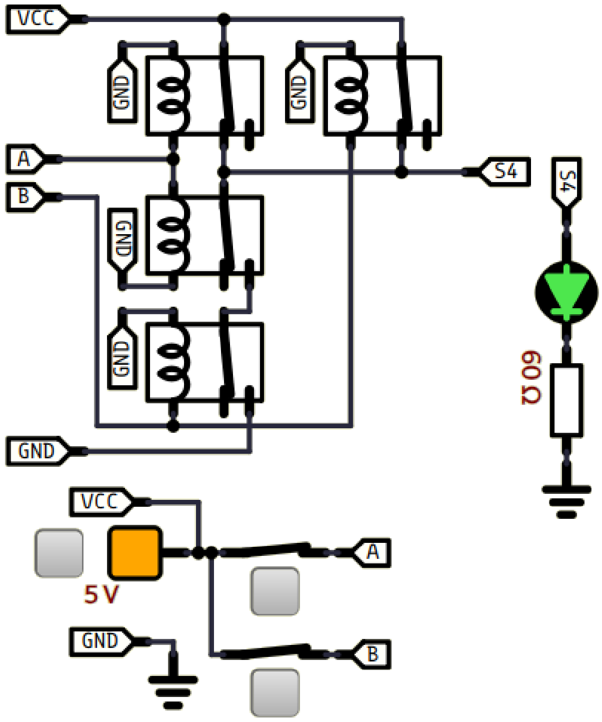
(simulide)




VCC
GND
A
B
S4
60 Ω
5 V


VCC
GND
A
B
S4
60 Ω
5 V


VCC
GND
A
B
S4
60 Ω
5 V

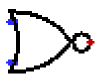

VCC
GND
A
B
S4
60 Ω
5 V

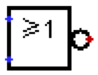
1-5) NON-OU (NOR) :

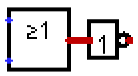
Transistors

(logisim)

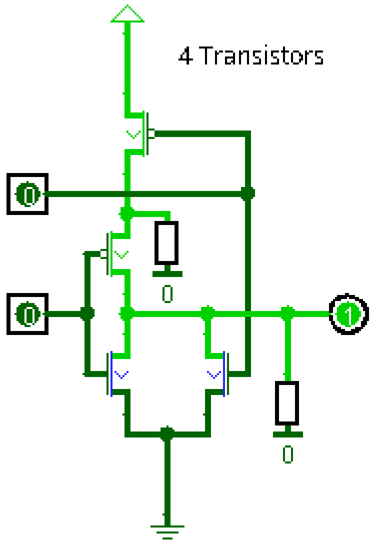
Américain Europe



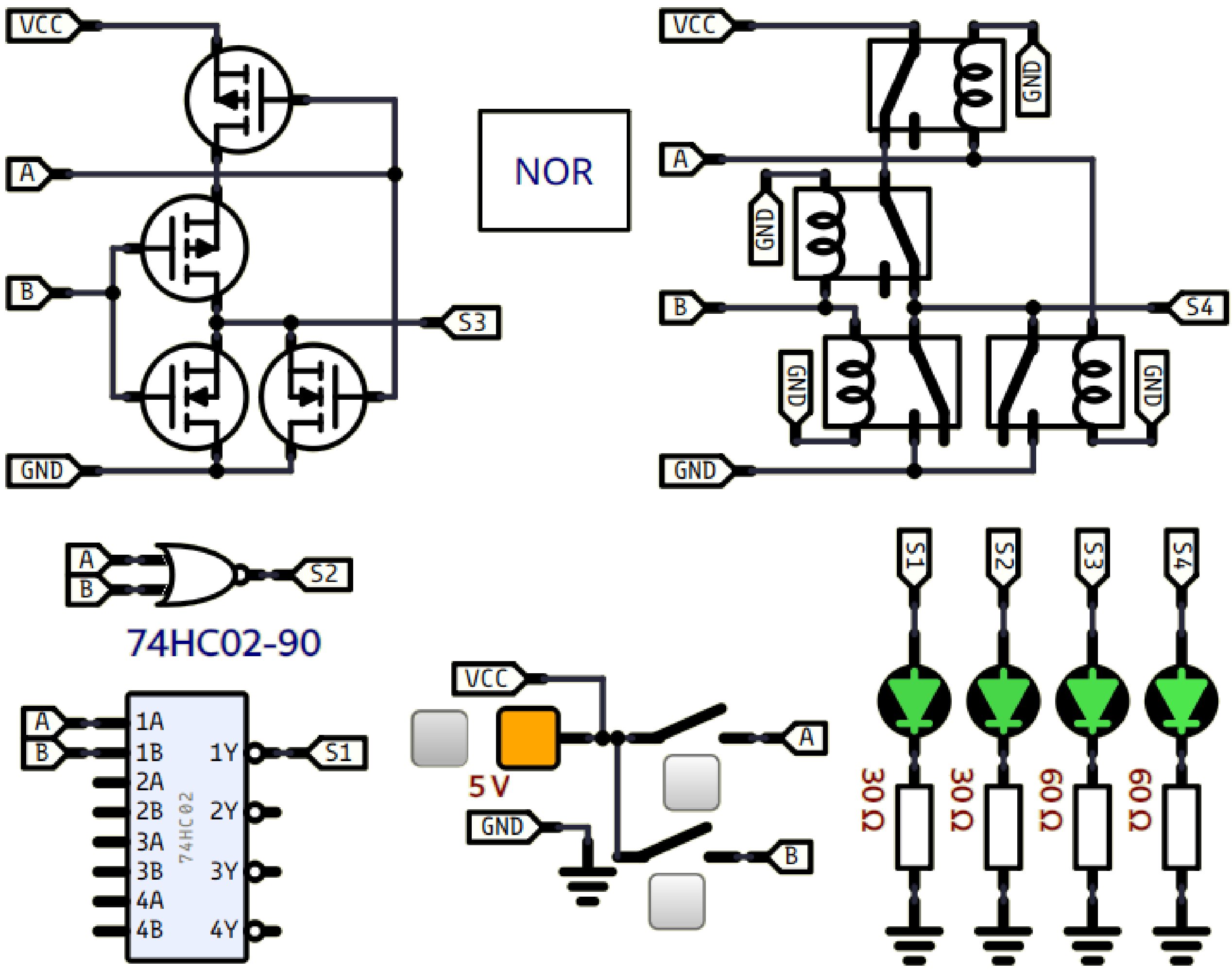


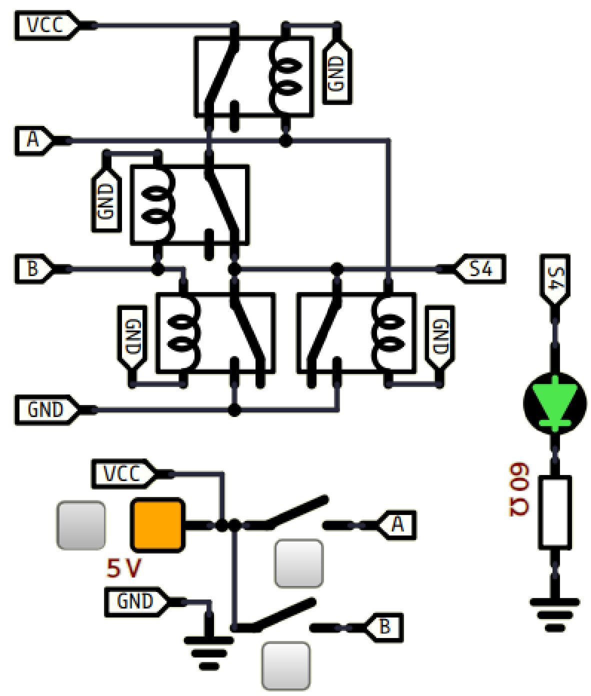


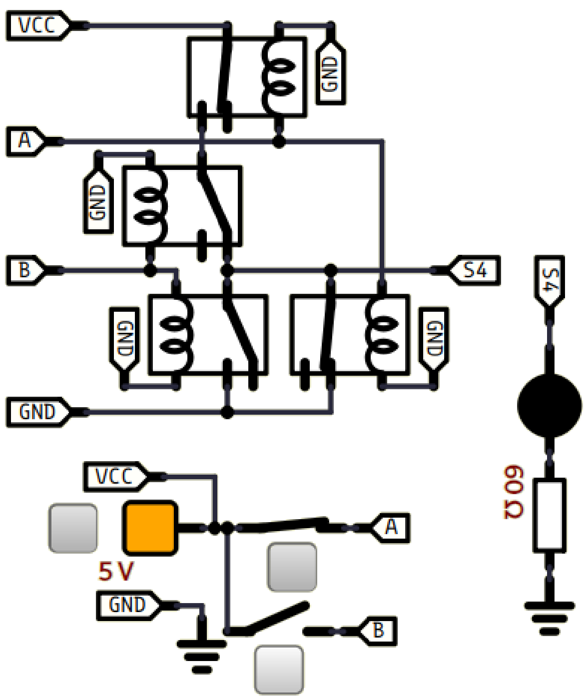
A	B	S
0	0	1
0	1	0
1	0	0
1	1	0

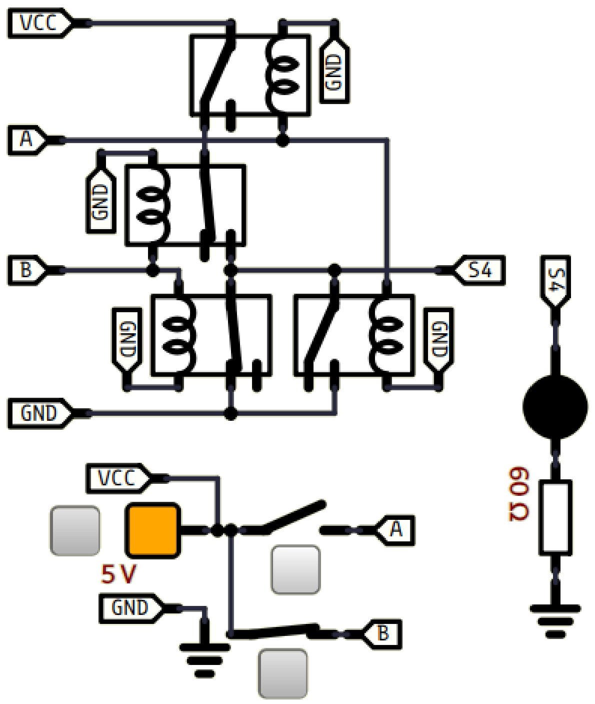


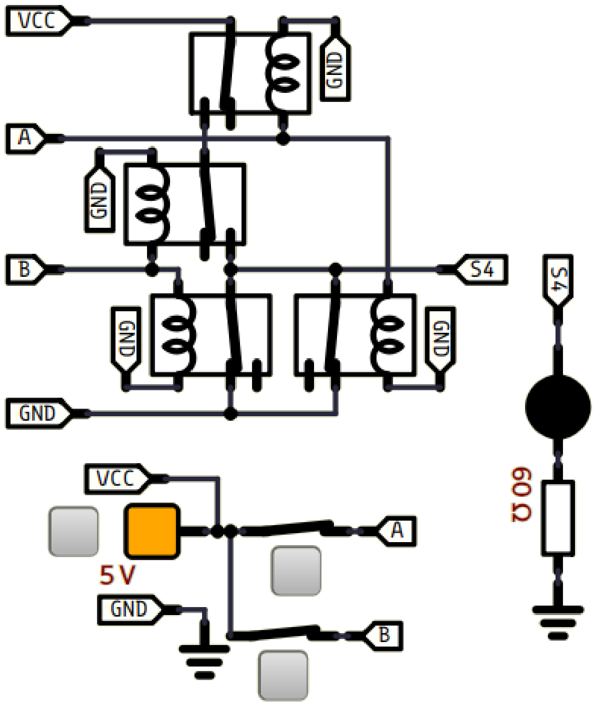
(simulide)




VCC
A
B
GND
S4
60 Ω
5 V

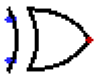

VCC
A
B
GND
S4
60 Ω
5 V


VCC
A
B
GND
S4
60 Ω
5 V

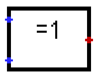

VCC
A
B
GND
S4
60 Ω
5 V

1-6) OU-EXCLUSIF (XOR) :

Américain



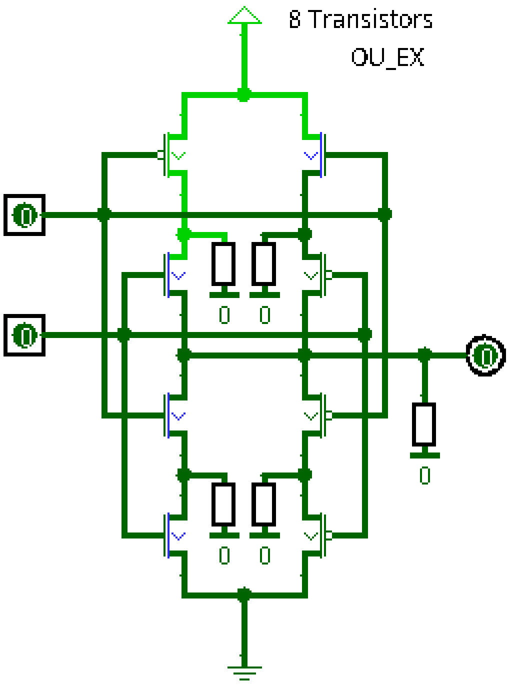
Europe

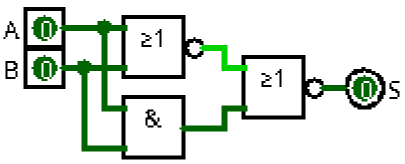


A	B	S
0	0	0
0	1	1
1	0	1
1	1	0

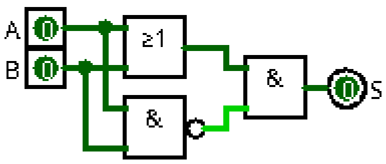
Transistors

(logisim)

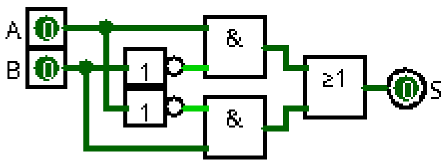




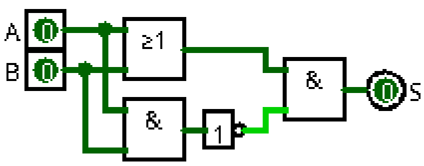
XOR => 3*4 = 12 Transistors



XOR => 3*4 = 12 Transistors

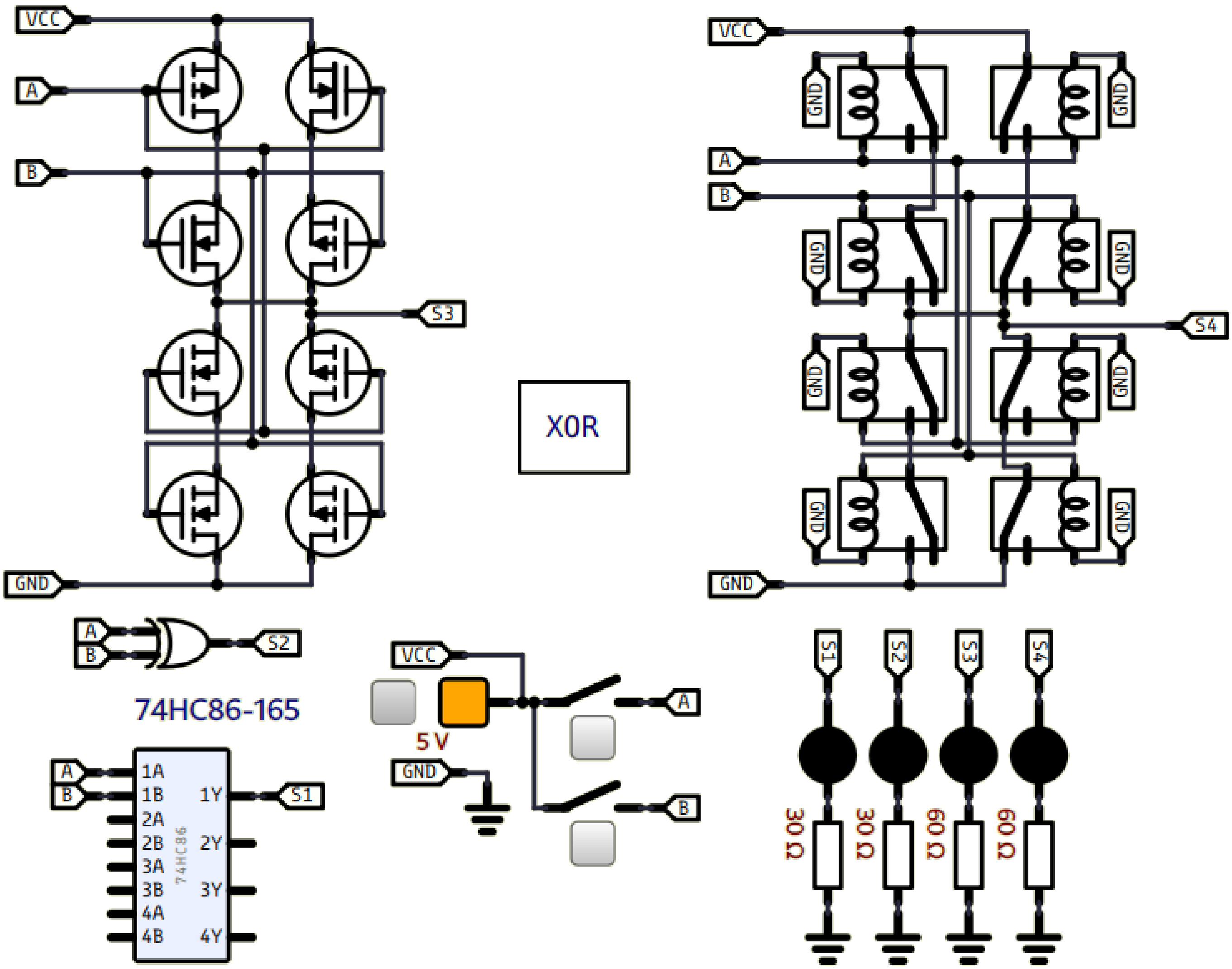


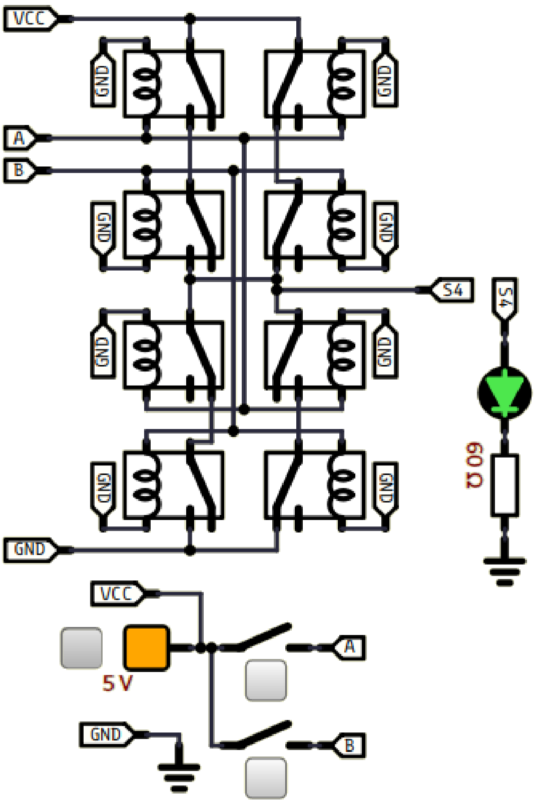
XOR => 3*4+2*2 = 16 Transistors

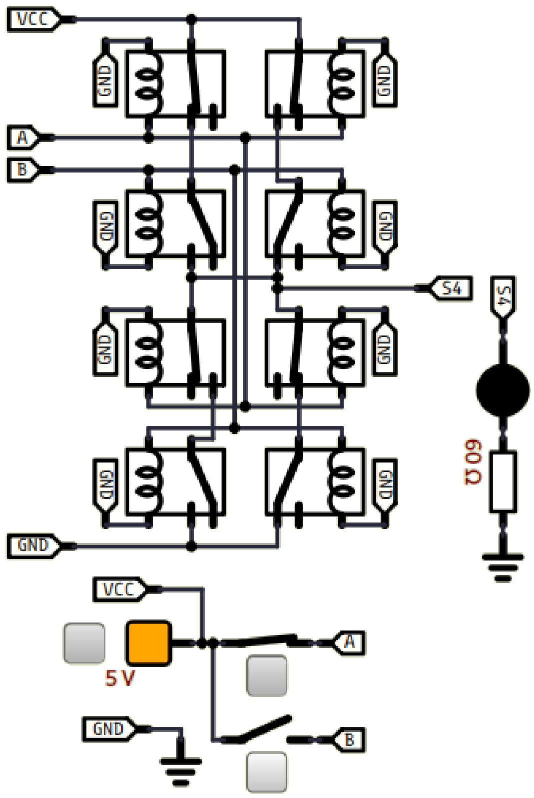


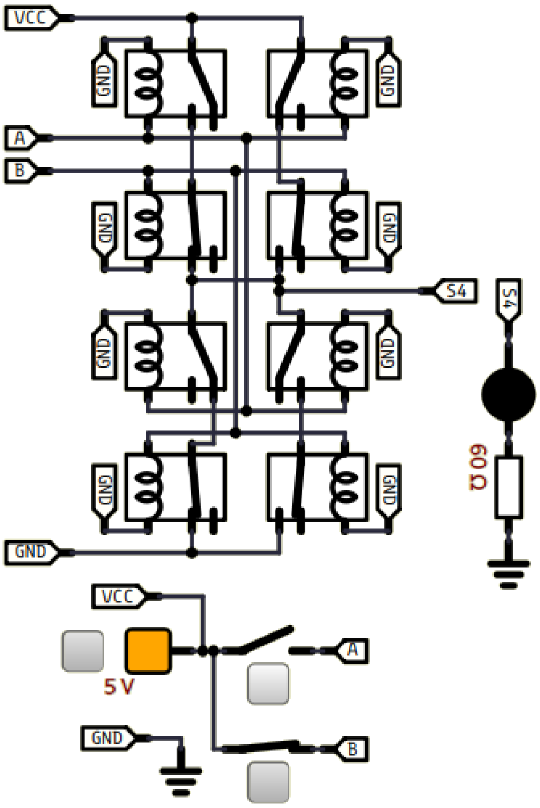
XOR => 3*4 + 2 = 14 Transistors

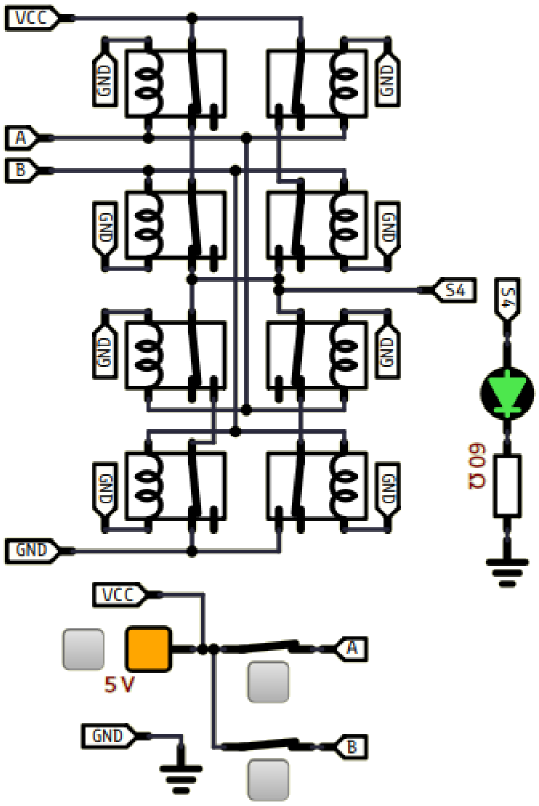
(simulide)


VCC
A
B
S3
GND
XOR
VCC
GND
A
B
S4
S2
74HC86-165
1A
1B
2A
2B
3A
3B
4A
4B
1Y
2Y
3Y
4Y
74HC86
S1
5 V
S1
S2
S3
S4
30 Ω
30 Ω
60 Ω
60 Ω


VCC
GND
A
B
S4
60 Ω
5 V

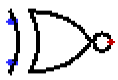

VCC
GND
A
B
S4
60 Ω
5 V


VCC
GND
A
B
S4
60 Ω
5 V

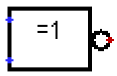

VCC
GND
A
B
S4
60 Ω
5 V

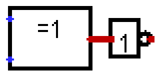
1-7) NON-OU-EXCLUSIF (XNOR) :

Américain



Europe

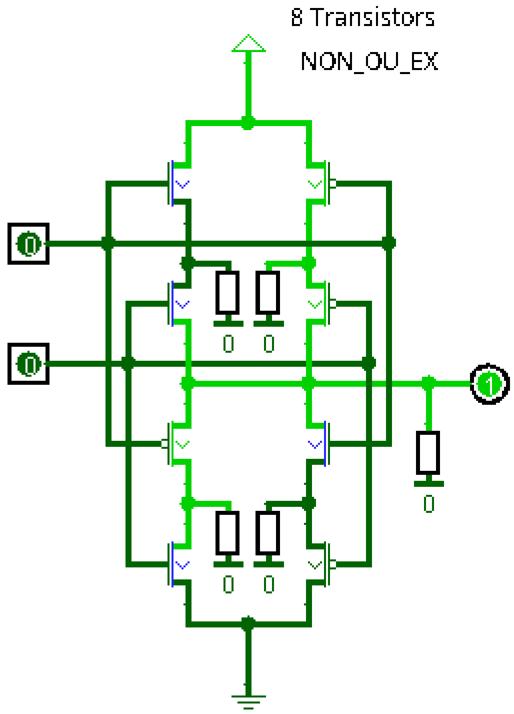


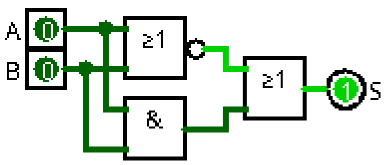


A	B	S
0	0	1
0	1	0
1	0	0
1	1	1

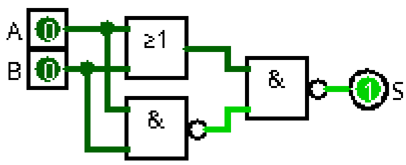
Transistors

(logisim)

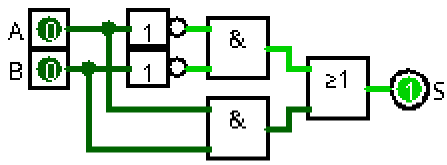




XOR => 3*4 = 12 Transistors

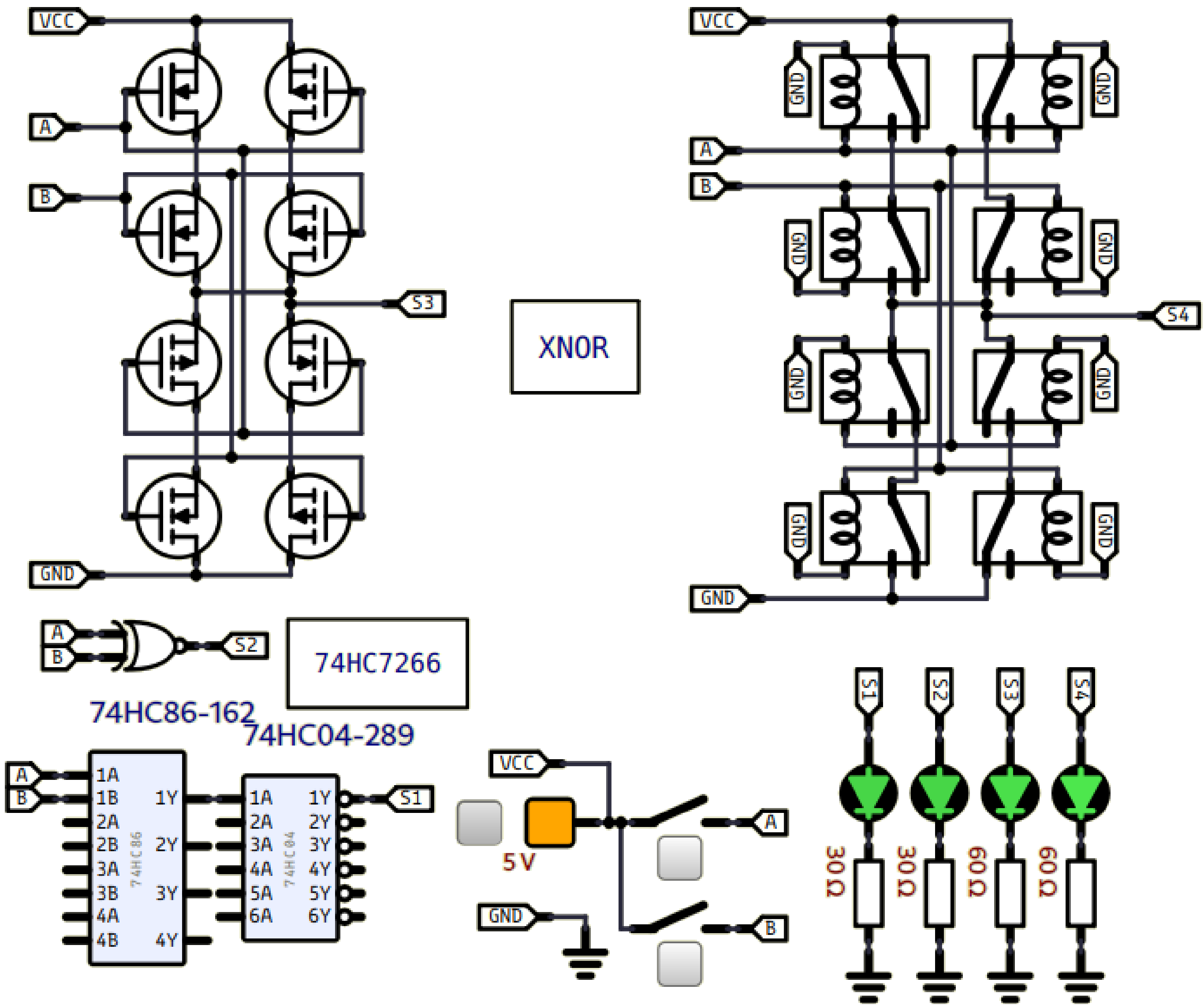


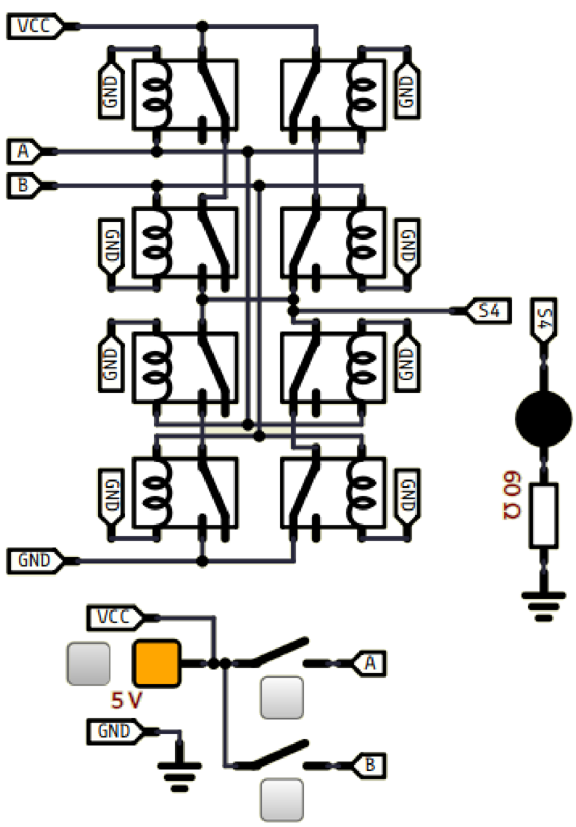
XOR => 3*4 = 12 Transistors

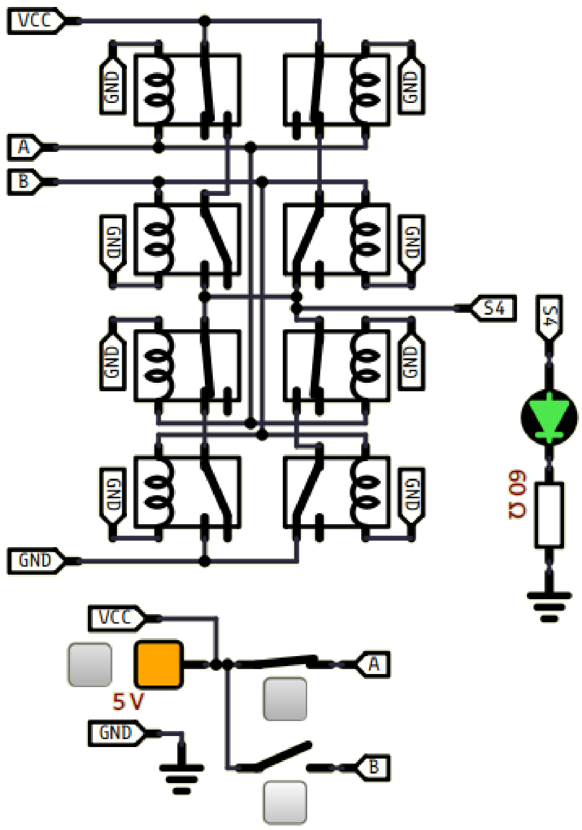


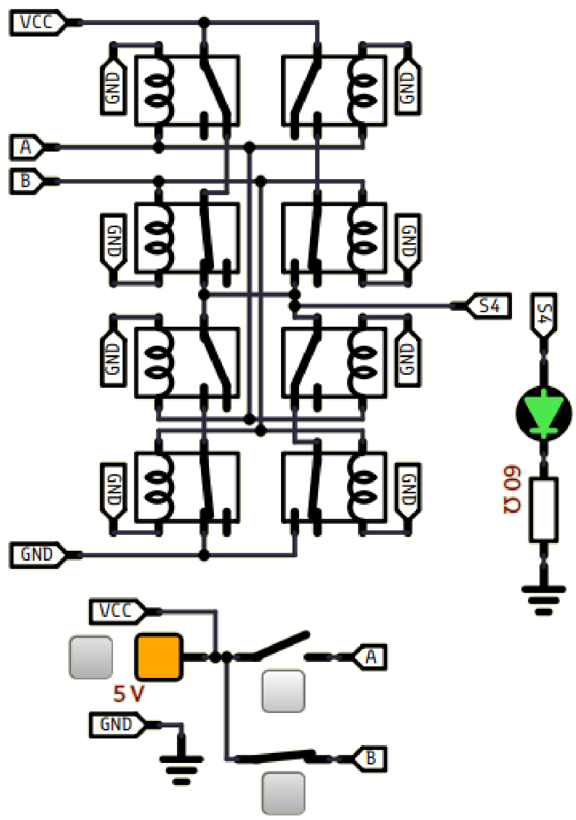
XOR => 3*4+2*2 = 16 Transistors

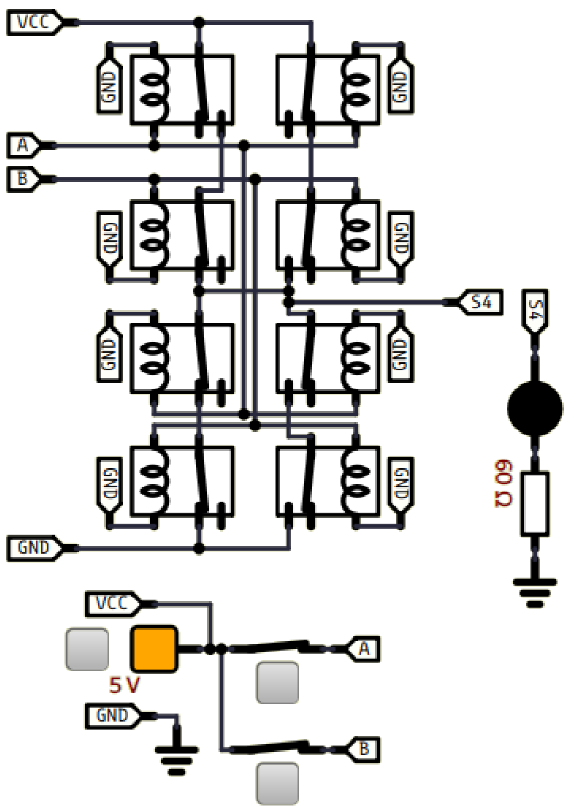
(simulide)


VCC
A
B
S3
GND
XNOR
S4
S2
74HC7266
74HC86-162
74HC04-289
1A
1B
1Y
2A
2B
2Y
3A
3B
3Y
4A
4B
4Y
S1
5 V
S1
S2
S3
S4
30 Ω
30 Ω
60 Ω
60 Ω


VCC
A
B
GND
S4
60 Ω
5 V

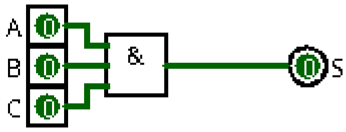

VCC
A
B
GND
S4
60 Ω
5 V

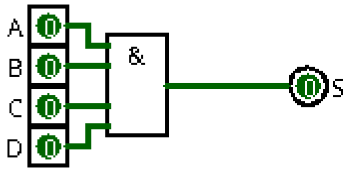

VCC
A
B
GND
S4
60 Ω
5 V

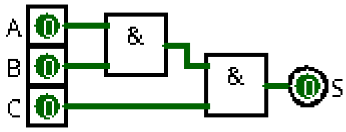

VCC
A
B
GND
S4
60 Ω
5 V

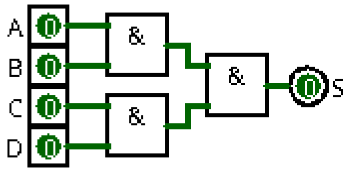
2) Portes logiques comportant plus de 2 entrées :

2-1) ET (AND) [3 à 4 entrées] :





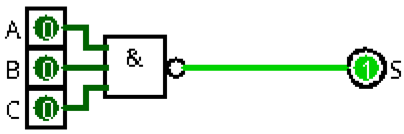


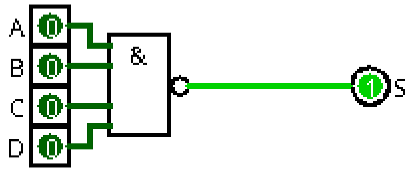


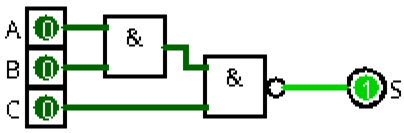
A	B	C	S
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

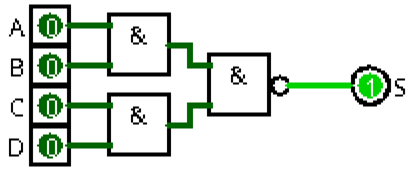
A	B	C	D	S
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

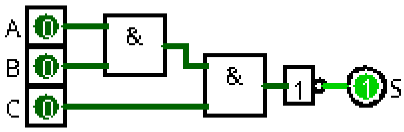
2-2) NON-ET (NAND) [3 à 4 entrées] :

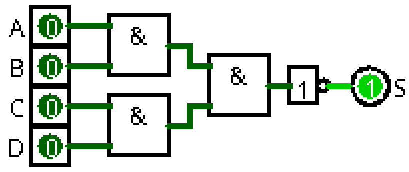








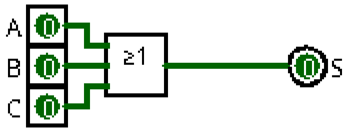


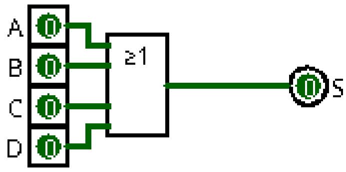


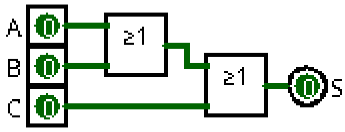
A	B	C	S
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

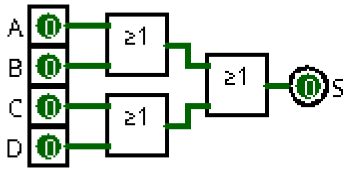
A	B	C	D	S
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

2-3) OU (OR) [3 à 4 entrées] :





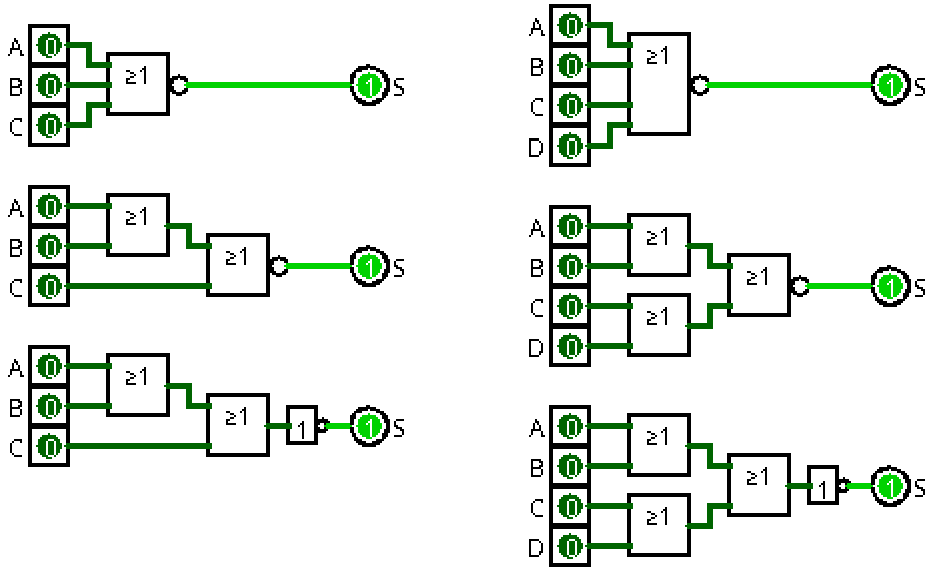




A	B	C	S
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

A	B	C	D	S
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

2-4) NON-OU (NOR) [3 à 4 entrées] :

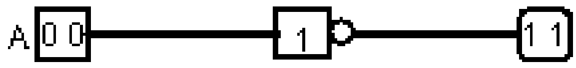


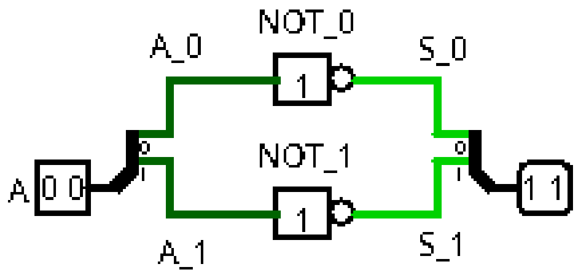
A	B	C	S
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

A	B	C	D	S
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

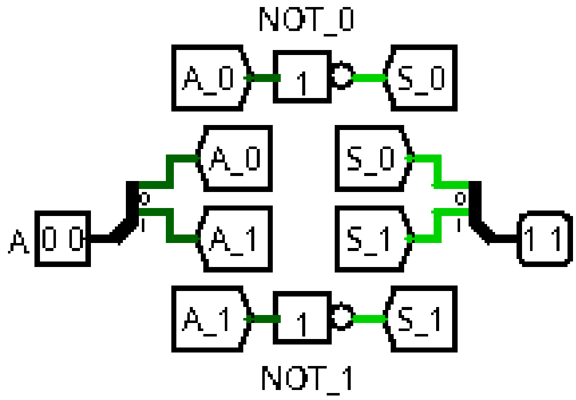
3) Portes logiques avec bus de données :

3-1) NON (NOT) {inverseur} :

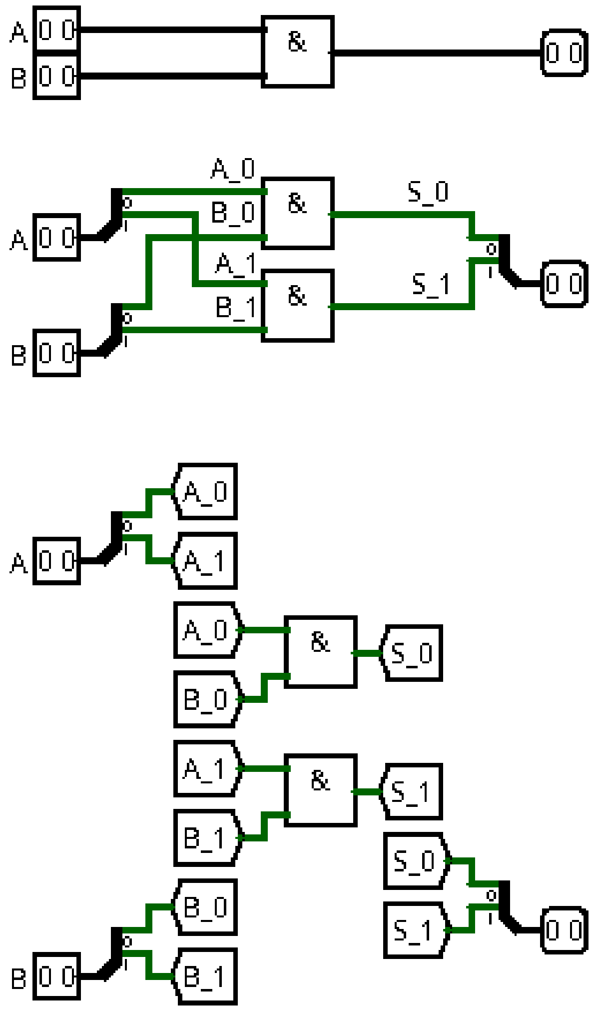




A		S	
0	0	1	1
0	1	1	0
1	0	0	1
1	1	0	0

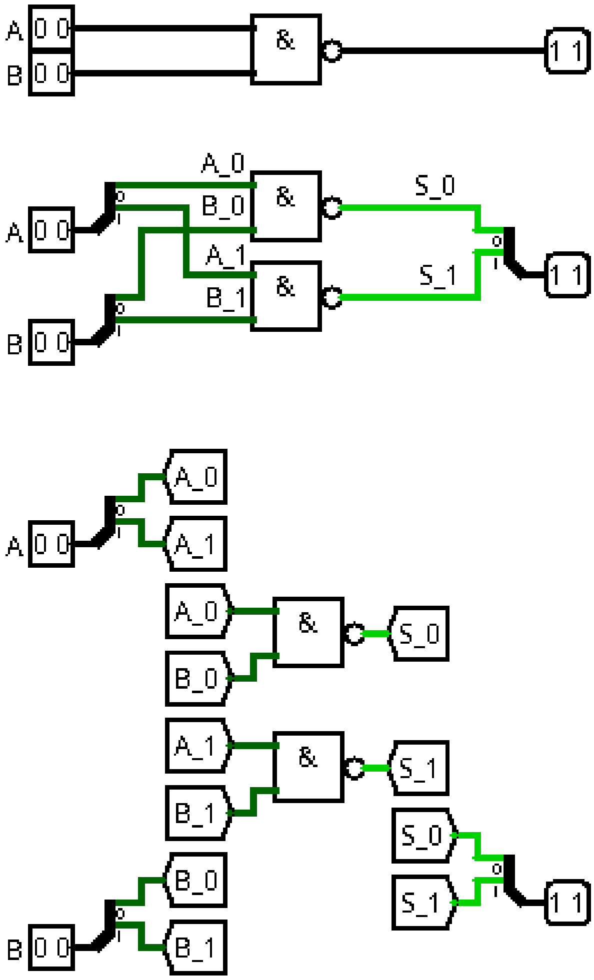


3-2) ET (AND) :



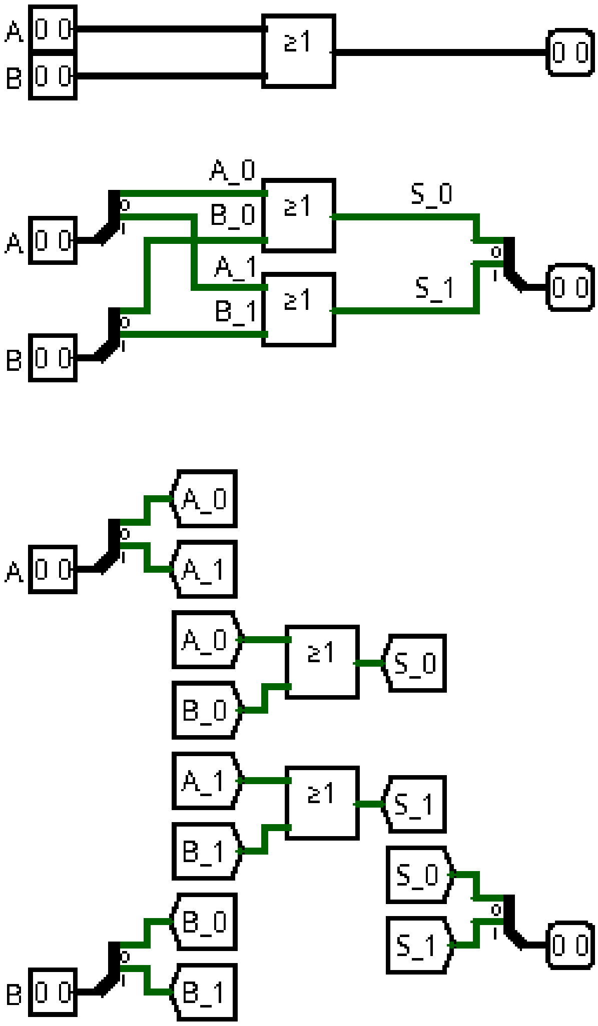
A		B		S	
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	0	0	0
0	0	1	1	0	0
0	1	0	0	0	0
0	1	0	1	0	1
0	1	1	0	0	0
0	1	1	1	0	1
1	0	0	0	0	0
1	0	0	1	0	0
1	0	1	0	1	0
1	0	1	1	1	0
1	1	0	0	0	0
1	1	0	1	0	1
1	1	1	0	1	0
1	1	1	1	1	1

1-3) NON-ET (NAND) :



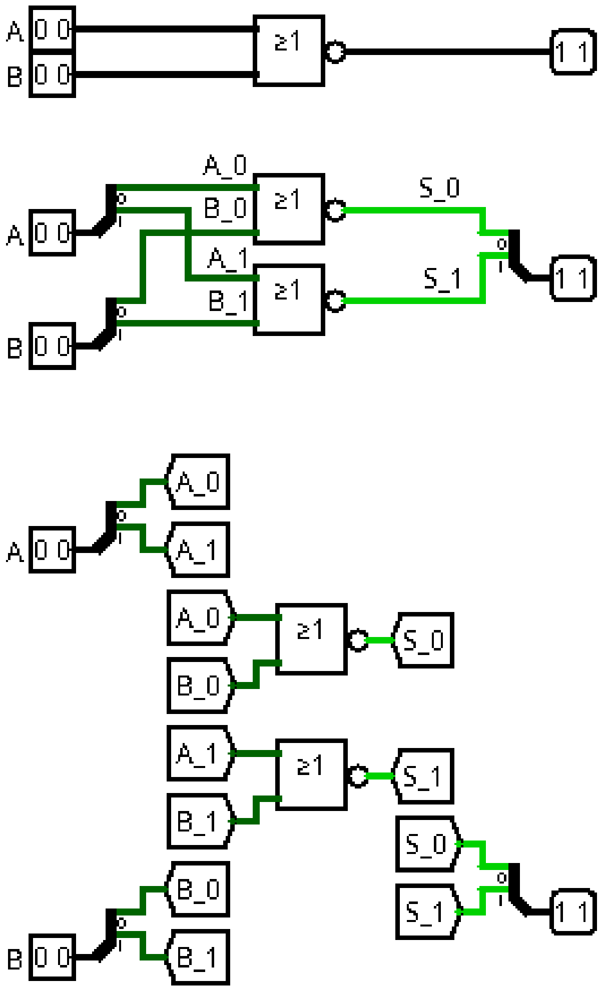
A		B		S	
0	0	0	0	1	1
0	0	0	1	1	1
0	0	1	0	1	1
0	0	1	1	1	1
0	1	0	0	1	1
0	1	0	1	1	0
0	1	1	0	1	1
0	1	1	1	1	0
1	0	0	0	1	1
1	0	0	1	1	1
1	0	1	0	0	1
1	0	1	1	0	1
1	1	0	0	1	1
1	1	0	1	1	0
1	1	1	0	0	1
1	1	1	1	0	0

3-4) OU (OR) :



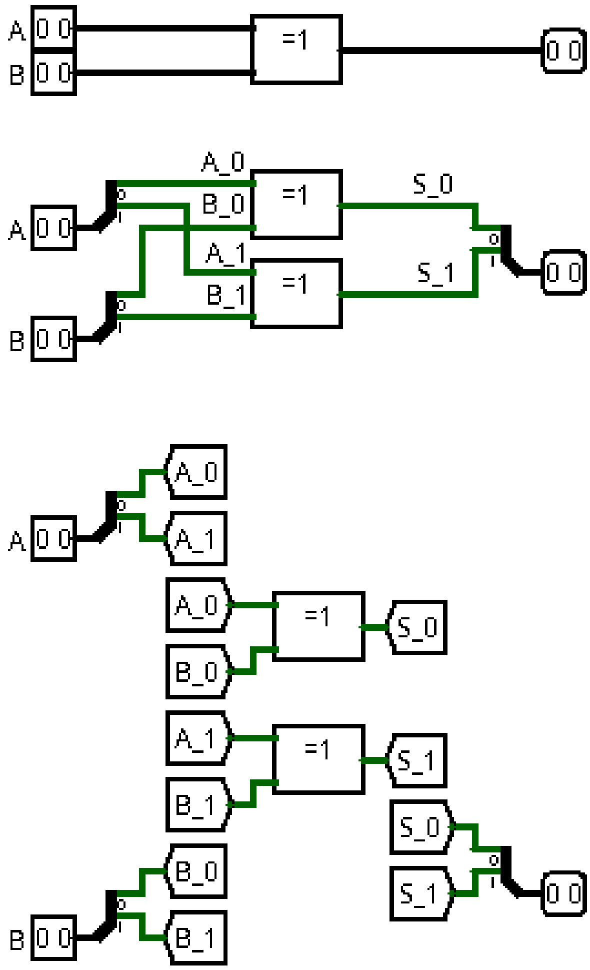
A		B		S	
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	1
0	1	0	0	0	1
0	1	0	1	0	1
0	1	1	0	1	1
0	1	1	1	1	1
1	0	0	0	1	0
1	0	0	1	1	1
1	0	1	0	1	0
1	0	1	1	1	1
1	1	0	0	1	1
1	1	0	1	1	1
1	1	1	0	1	1
1	1	1	1	1	1

3-5) NON-OU (NOR) :



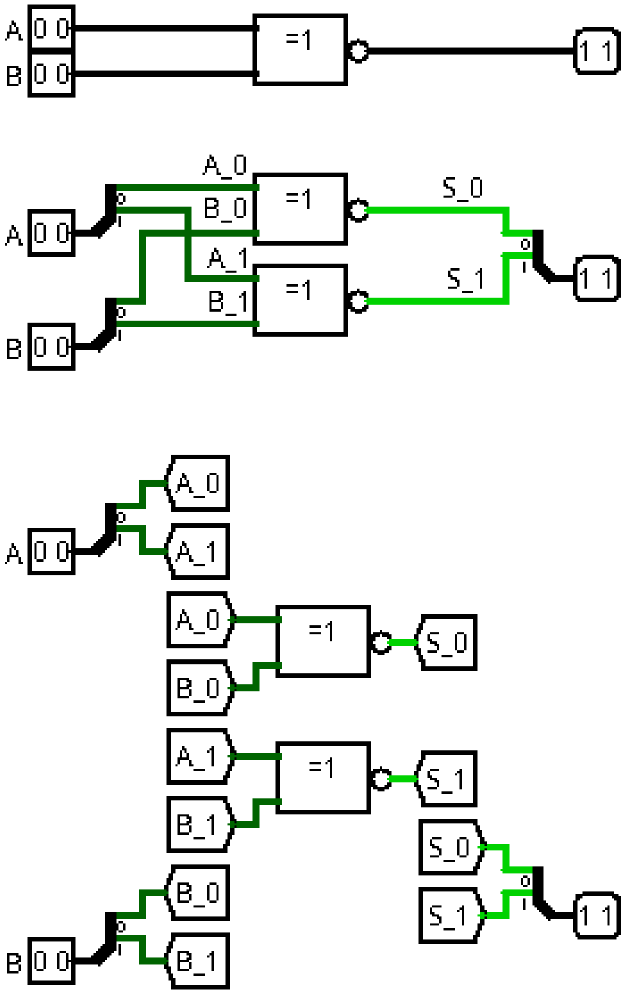
A		B		S	
0	0	0	0	1	1
0	0	0	1	1	0
0	0	1	0	0	1
0	0	1	1	0	0
0	1	0	0	1	0
0	1	0	1	1	0
0	1	1	0	0	0
0	1	1	1	0	0
1	0	0	0	0	1
1	0	0	1	0	0
1	0	1	0	0	1
1	0	1	1	0	0
1	1	0	0	0	0
1	1	0	1	0	0
1	1	1	0	0	0
1	1	1	1	0	0

3-6) OU-EXCLUSIF (XOR) :



A		B		S	
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	1
0	1	0	0	0	1
0	1	0	1	0	0
0	1	1	0	1	1
0	1	1	1	1	0
1	0	0	0	1	0
1	0	0	1	1	1
1	0	1	0	0	0
1	0	1	1	0	1
1	1	0	0	1	1
1	1	0	1	1	0
1	1	1	0	0	1
1	1	1	1	0	0

3-7) NON-OU-EXCLUSIF (XNOR) :



A		B		S	
0	0	0	0	1	1
0	0	0	1	1	0
0	0	1	0	0	1
0	0	1	1	0	0
0	1	0	0	1	0
0	1	0	1	1	1
0	1	1	0	0	0
0	1	1	1	0	1
1	0	0	0	0	1
1	0	0	1	0	0
1	0	1	0	1	1
1	0	1	1	1	0
1	1	0	0	0	0
1	1	0	1	0	1
1	1	1	0	1	0
1	1	1	1	1	1